- $\forall k \in \mathbb{N}$, a has no effect on cov

$$\begin{aligned}\text{Cov}(Y_t, Y_{t-k}) &= \text{Cov}(bZ_t + cZ_{t-1}, bZ_{t-k} + cZ_{t-k-1}) \\ &= b^2\text{Cov}(Z_t, Z_{t-k}) + \overbrace{bc\,\text{Cov}(Z_t, Z_{t-k-1})}^{0} \\ &\quad + cb\,\text{Cov}(Z_{t-1}, Z_{t-k}) + c^2\text{Cov}(Z_{t-1}, Z_{t-k-1})\end{aligned}$$

$$\text{Cov}(Y_t, Y_{t-k}) = \begin{cases} (b^2 + c^2)\sigma^2, & \text{if } k = 0 \\ cb\sigma^2, & \text{if } k = 1 \\ 0, & \text{else} \end{cases}$$

$\gamma_Y(k)$ is only a function of the lag k

We can conclude that $\{Y_t\}$ is stationary.